Findings of the 2009 Workshop on Statistical Machine Translation

Summary of the paper by Callison-Burch, et al. (2009)

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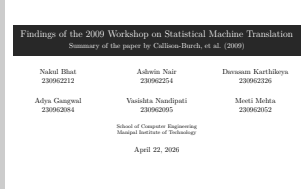
School of Computer Engineering
Manipal Institute of Technology

April 22, 2026

2026-04-20

WSMT 2009



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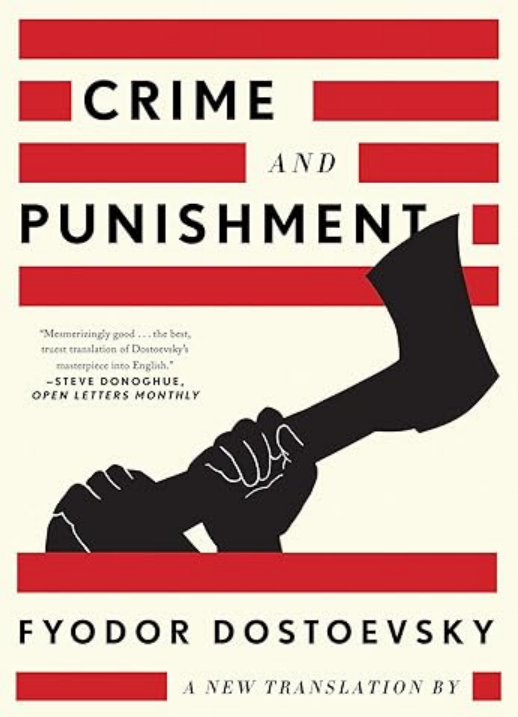
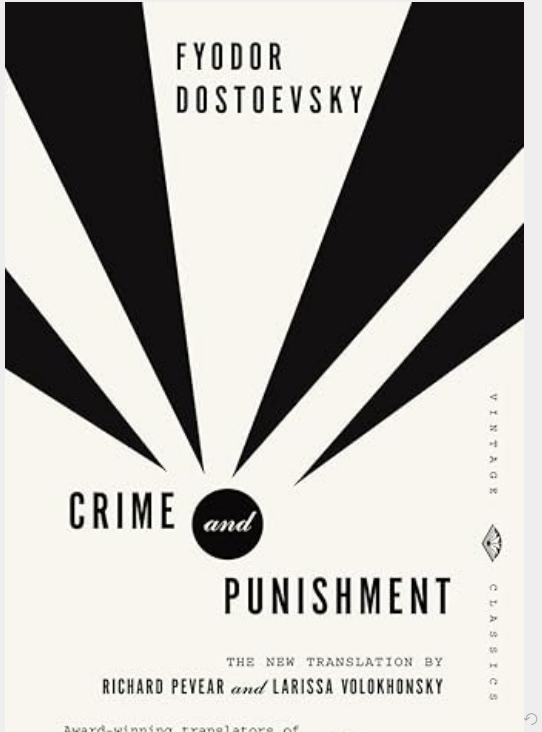

CRIME
AND
PUNISHMENT
"Mesmerizingly good . . . the best, truest translation of Dostoevsky's masterpiece into English."
–STEVE DONOGHUE, OPEN LETTERS MONTHLY
FYODOR DOSTOEVSKY
A NEW TRANSLATION BY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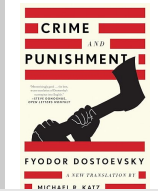
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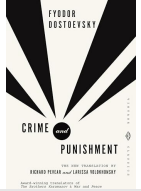
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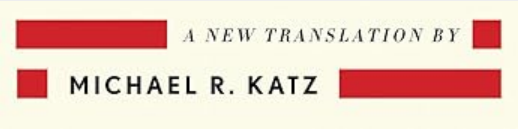
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WMT 2009 Overview

- Fourth Workshop on Statistical Machine Translation
- Shared task evaluation of machine translation systems

Goals of the Workshop

- Evaluate state-of-the-art machine translation
- Provide shared datasets and benchmarks
- Compare systems under common conditions
- Improve evaluation methodologies

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Translation Task

- News domain translation
- English and five European languages

Training Data	Testing Data
Automated web crawler data, and existing datasets	Manually translated news articles.

Datasets Used

Translation Datasets

- Europarl Translation Corpus
- News Commentary Translation Corpus
- 10^9 Word Parallel Corpus

Language Datasets

- Europarl Language Model Corpus
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System Combination Task

- Participants were allowed to use **Moses**, an open-source translation toolkit.
- Moses performed at par with the best submissions of the previous year.
- 22 groups from 20 institutions submitted; 12 groups used the Moses toolkit.
- 87 systems were submitted, and an additional 42 were submitted as secondary systems. 6 combination systems were also submitted.

System Combination

- Naïve Combination
- Multi-source combination

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Naïve Combination
Multi-source combination

Evaluation Task

- Outputs from select systems were evaluated by human annotators. These were split into two types:
  1. Ranked lists
  2. Edited translations
- Automated metrics also scored the translations.
- Both scores were compared to find the best automated metrics.

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Next Presenter: Ashwin Nair

Human Evaluation

Ashwin Nair

Human Evaluation

Ashwin Nair

Statistics on Human Effort

- Manual evaluation done by:-
  - shared task participant
  - interested volunteers
  - small number of paid of annotators

Participation Statistics

- More than 160 people participated, with 100 people putting more than an hours worth of work, and 30 putting more than 4 hours.
- collectively 479 hours was spent.

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Overview of Evaluation Strategy

Core Philosophy

Automatic metrics are imperfect, greater emphasis on human judgment

Three Primary Tasks

- Sentence Ranking: The official determinant of system quality.
- Editing MT Output: Assessing fluency and intelligibility.
- Acceptability Judging: Verifying the correctness of edited output.

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Task 1 — Sentence Ranking

The Official Metric

- The Process: Annotators are shown a source sentence and up to 5 system translations.
- The Instruction: "Rank translations from Best to Worst (ties allowed)."
- Scoring: Systems are ranked based on how frequently they are judged "better than or equal to" others.

Scale

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Task 2 — Editing MT Output

Measuring Fluency and Intelligibility

- Experimental Design: Judges see MT output without the source or reference translation.
- The Task: Edit the sentence to be as fluent as possible.

Three Outcomes

- Edited: Sentence was understandable but needed polish.
- No corrections needed: Sentence was already perfect.
- Unable to correct: MT output was too garbled to understand.

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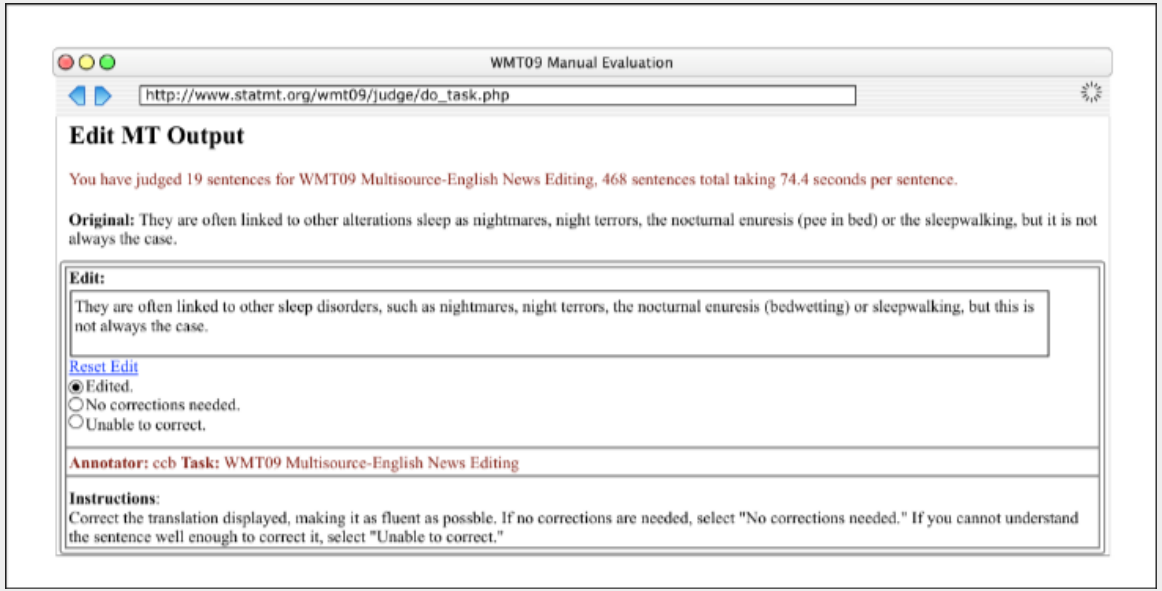

Figure: Interface for Editing MT Output

Task 2 — Editing MT Output

Figure: Interface for Editing MT Output

Task 3 — Judging Acceptability

Validation of the Edits

- The Setup: A second group of judges reviews the edited sentences alongside the original reference.
- The Question: Is this edited version a ”fully fluent and meaning-equivalent” alternative to the reference?

Outcome

A simple Yes/No indicator for over 31,000 edited items.

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WMT09 Manual Evaluation

http://www.statmt.org/wmt09/judge/do_task.php

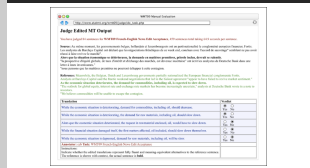
Judge Edited MT Output

You have judged 84 sentences for **WMT09 French-English News Edit Acceptance**, 459 sentences total taking 64.9 seconds per sentence.

Source: Au même moment, les gouvernements belges, hollandais et luxembourgeois ont en parti nationalisé le conglomérat européen financier, Fortis. Les analystes de Barclays Capital ont déclaré que les négociations frénétiques de ce week end, conclues avec l'accord de sauvetage" semblent ne pas avoir réussi à faire revivre le marché".
Alors que la situation économique se détériorasse, la demande en matières premières, pétrole inclus, devrait se ralentir.
"la prospective d'équité globale, de taux d'intérêt et d'échange des marchés, est devenue incertaine" ont écrit les analystes de Deutsche Bank dans une lettre à leurs investisseurs."
"nous pensons que les matières premières ne pourront échapper à cette contagion.

Reference: Meanwhile, the Belgian, Dutch and Luxembourg governments partially nationalized the European financial conglomerate Fortis.
Analysts at Barclays Capital said the frantic weekend negotiations that led to the bailout agreement "appear to have failed to revive market sentiment."
As the economic situation deteriorates, the demand for commodities, including oil, is expected to slow down.
"The outlook for global equity, interest rate and exchange rate markets has become increasingly uncertain," analysts at Deutsche Bank wrote in a note to investors.
"We believe commodities will be unable to escape the contagion.

Translation	Verdict
While the economic situation is deteriorating, demand for commodities, including oil, should decrease.	(●) Yes ( ) No
While the economic situation is deteriorating, the demand for raw materials, including oil, should slow down.	(●) Yes ( ) No
Alors que the economic situation deteriorated, the request in rawmaterial enclosed, oil, would have to slow down.	( ) Yes (●) No
While the financial situation damaged itself, the first matters affected, oil included, should slow down themselves.	( ) Yes (●) No
While the economic situation is depressed, demand for raw materials, including oil, will be slow.	( ) Yes (●) No
Annotator: ccb **Task:** WMT09 French-English News Edit Acceptance	
Instructions: Indicate whether the edited translations represent fully fluent and meaning-equivalent alternatives to the reference sentence. The reference is shown with context, the actual sentence is **bold**.	



Quality Control – Annotator Agreement

Measuring Reliability (Kappa Coefficient K)

$$K = (P(A) - P(E))(1 - P(E))$$

- P(A) - the proportion of times that the annotators agree
- P(E) - the proportion of time that they would agree by chance.

- Intra-annotator agreement: How often a judge agrees with themselves (Substantial: K = 0.56 – 0.73).
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Optimization

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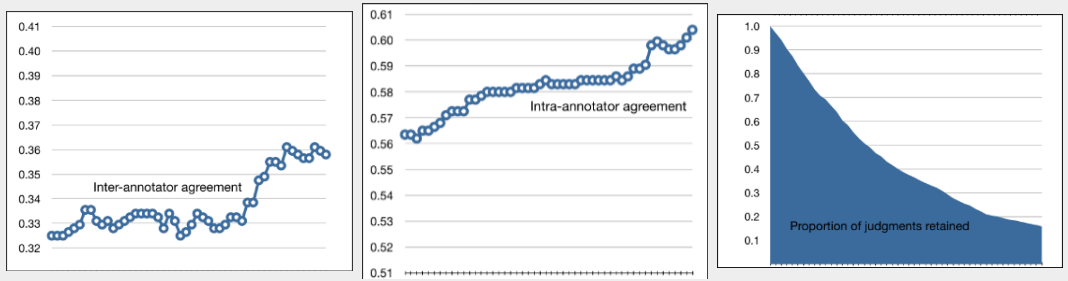
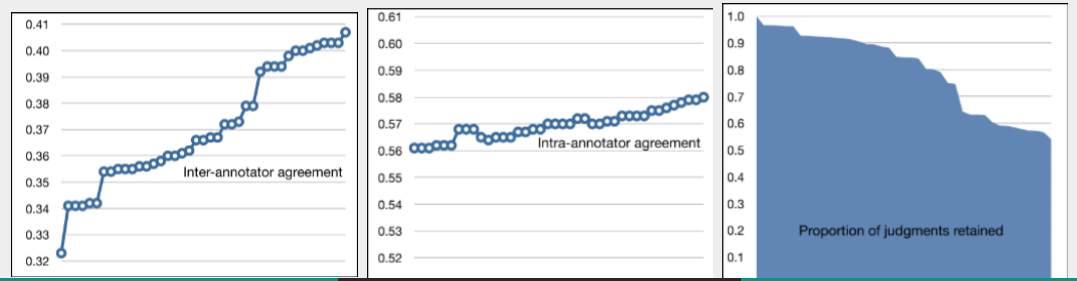
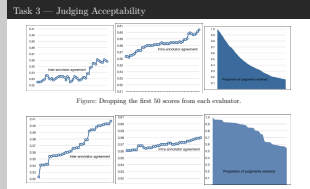


Figure: Dropping the first 50 scores from each evaluator.





Key Takeaways

- Human-Centric: WMT09 prioritizes human ranking over BLEU scores to validate systems.
- Scale: Massive crowdsourced effort from participants, volunteers, and paid staff.
- Innovation: Introduced "editing-based" evaluation to move beyond simple ranking.
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Next Presenter: Davasam Karthikeya

Translation Results

Davasam Karthikeya

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Translation Task Results

In this section, we will try to answer these three questions:

- Which systems produced the best translation quality for each language pair?
- Did the system combinations produce better translations than individual systems?
- Which of the systems that used only the provided training materials produced the best translation quality?

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Which MT systems are best?

How "Best" is Defined?

- Based on human evaluation (sentence ranking)
- Systems ranked by:
- How often their translations are better than or equal to others
- They used Sign Test ($p \leq 0.1$)

Pair	Performance Range	Top systems (C)	Top systems (NC)
French → English	0.76 - 0.34	DCU, JHU	Google, LIMSI
English → French	0.73 - 0.38	UKA, DCU	LIUM-SYSTRAN, Google

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German → English	0.66 - 0.33	STUTTGART, UEDIN	RBMT5, USAAR
English → German	0.66 - 0.43	UKA, UEDIN	RBMT2, RBMT3
Czech → English	0.75 - 0.51	UEDIN, CU-BOJAR	Google
Hungarian → English	0.75 - 0.45	UMD, UEDIN	MORPHO

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System Combinations vs Individual Systems

Language Pair	Best system combinations	Entries	Significantly different than best individual systems?
German-English	RWTH-COMBO, BBN-COMBO, CMU-COMBO, USAAR-COMBO	5	BBN-COMBO > GOOGLE, SYSTRAN; USAAR-COMBO < RMBT2; no difference for others
English-German	USAAR-COMBO	1	worse than 3 best systems
Spanish-English	CMU-COMBO, USAAR-COMBO, BBN-COMBO	3	each better than one of the RBMT systems, but there was no difference with GOOGLE, TALP-UPC
English-Spanish	USAAR-COMBO	1	no difference
French-English	CMU-COMBO-HYPOSEL, DCU-COMBO, CMU-COMBO	5	no difference
English-French	USAAR-COMBO, DCU-COMBO	2	USAAR-COMBO > UKA; DCU-COMBO > SYSTRAN, LIMSI; no difference with others
Czech-English	CMU-COMBO	2	no difference
Hungarian-English	CMU-COMBO-HYPOSEL, CMU-COMBO	3	both worse than MORPHO
Multisource-English	RWTH-COMBO	3	N/A

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Are system combinations better than single systems?

Understandability

- Measures how often translated output is clear and meaningful after editing
- Based on human judgment (Yes/No) on edited translations
- Reflects real-world usability of translations

Results:

- 50% acceptable translations (French - English, German - English, English - Spanish)
- 40% acceptable translations (Spanish - English)
- 30% acceptable translations (English - Czech)
- 20% acceptable translations (Czech - English, Hungarian - English)

- Even top systems are only correct half the time
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- Commercial systems (Google, RBMT) perform strongly
- Performance drops for complex/hard language pairs
- Many systems perform similarly → hard to declare a clear winner
- System combinations doesn't guarantee improvement
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Next Presenter: Adya Gangwal

Shared Evaluation

Adya Gangwal

Shared Evaluation Task Overview

- Goal: Evaluate automatic MT evaluation metrics
- Compare metrics against human judgments
- Builds on previous Metrics-MATR evaluation (NIST)

Table: Automatic Metrics Evaluated by Type

Metric Type	Specific Metrics Evaluated
N-gram based	BLEU, METEOR
Edit-distance	TER, TERp
Similarity-based	MaxSim
Semantic	RTE, SemPOS
Combined	ULC
POS-based	wpF, wpBLEU

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System-Level Correlation

- Compare metric rankings with human rankings
- Use Spearman rank correlation (ρ)

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

where,

d_i Difference between the rank for system i

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- Compare metrics at sentence level
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Next Presenter: Vasishta Nandipati

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Vasishta Nandipati

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Spearman's rank correlation (ρ).
$\rho = 1$ means perfect match with humans.
$\rho =$ -1 means perfect reverse match with humans.

Results

- For non-English targets: metrics relying on English tools (WordNet, POS taggers) fail. TERp and wpBleu are best there.
- English-German: nearly all metrics had negative correlation.

Metric	Avg rho
ULC	0.83
MaxSim	0.80
RTE (absolute)	0.79
Meteor-rank	0.75
BLEU	0.47
TER / TERp	negative

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Method

for every pair of systems on one sentence, did the metric agree with the human? Random baseline = 50%.

Most metrics barely beat – or fall below – 50%.

Into English best

Metric	Score
ULC	0.54
RTE	0.53
wpF	0.52

Out of English best

Metric	Score
wpF	0.51
wcd6p4er	0.50

- TER / TERp: fell BELOW 50% – worse than random.

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Sentence-Level Consistency

The “prior” trick

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Next Presenter: Meeti Mehta

Conclusion

Meeti Mehta

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- 7 commercial systems and Google's online statistical MT system
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